



ReconScout XT micro-bot used in warfare, and to fight terrorism (Credit: www.designfax.net)

have to surmount the challenge of overcoming non-linear physics at this miniature scale.

Forces and effects at small scales are different from those that are experienced at the normal scale. Gravity and inertial forces become less important or even negligible, and on the other side, surface forces or friction forces have striking influence on micro-bot behaviour. The mechanism of this influence is still not fully-understood, and more research is needed to develop complete understanding of the behaviour of micro-bots.

Other challenges are application-specific. For example, in industrial high-precision micro-robotics the challenge is development of fully-automated micro-bot cells for high-throughput operation using feedback from a scanning electron microscope (SEM). The bottleneck faced is the speed of SEM imaging.

Serious challenges faced in targeted medicine utilising mobile micro-bots are to achieve their predictable behaviour and controllability, biocompatibility and non-toxicity. Untethered micro-bots for monitoring applications, both civilian and

defence, require reliable and long-lasting power supply.

These applications involve significant challenges related to the design and communication in micro-bot swarms, depending on environment and application domain, such as flying, crawling or swimming.

Endnote

Research in MEMS and nano-technology progressing relentlessly will result in these devices becoming available at lower costs with a higher level of sophistication by combining sensors, processing power, a type of locomotion, and the method of storing or harvesting energy. One can look forward to medical robots becoming available, having the ability to chart one's unique physiologies, clean arteries, destroy kidney stones, assist in diagnosis and combating disease, and even fight cancer.

Industrial micro-bots will have the ability to monitor and repair machinery and infrastructure, and may become another tool for manufacturing.

Micro-bots will find use in warfare too, to attack enemies and destroy their weapons, equipment and manufacturing infrastructure, thus reducing their effectiveness.

A day will come when micro-bots would become an integral part of our lives as cellphones are now. However, the danger lies in the multitudes of nearly invisible robots being used by employers, or the power to log and record employees' activities, seriously affecting their privacy.

The future of micro-robotics is highly optimistic due to the ever-increasing interest of professionals in this comparatively new field. Rapid developments are taking place in many application domains. The coming years should see significant improvements in the performance of micro-bots and an increasing number of innovations. One thing is clear, our life will be more and more influenced by robotic technologies in which micro-robotics will play a significant role. **EFY**



A typical scanning electron microscope instrument used in micro-robotics (Credit: <https://serc.carleton.edu>)



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